

The multiplier variety is a Dirichlet curve exactly in rational rank one

Solution packet for arXiv:2507.21537v3

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Status. Candidate full solution, likely valid, awaiting human review.

1 Source problem

Ahmed, Das, and Sahu study the multiplier variety associated with a complete Nevanlinna–Pick Dirichlet-series kernel [1]. For positive weights $\mathbf{b} = (b_j)_{j=1}^d$ and distinct integers $\mathbf{n} = (n_j)_{j=1}^d$, where $d \in \mathbb{N}_{\geq 2} \cup \{\infty\}$, put

$$f_{\mathbf{b},\mathbf{n}}(s) = (\sqrt{b_j} n_j^{-s})_{j=1}^d.$$

Their Corollary 4.11 states that if

$$\{\log n_1, \log n_i\} \text{ is linearly dependent over } \mathbb{Q} \quad (i \geq 2), \quad (1)$$

then the associated multiplier variety V is exactly the image of the relevant half-plane, together with the origin. Its two alternatives are shown below.

$$V = \{ \bigwedge_{j=1}^d \log n_j \in \mathbb{R}^d : \exists \mathbf{u} \in \mathbb{R}^d \text{ such that } \mathbf{u} \cdot \mathbf{v} \leq 0 \text{ for all } \mathbf{v} \in V \}.$$

If V is the multiplier variety of a CNP kernel $K_{\mathbf{b},\mathbf{n}}$ defined on $\mathbb{H}_\rho$, then by Remark 4.1, the zero vector is always contained in V , and from the definition of multiplier variety (4.2) $f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_\rho) \subseteq V$. Thus, $\{0\} \cup f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_\rho) \subseteq V$. As an application of Theorem 4.6, the following result provides a family of CNP Dirichlet series kernels for which there is not much difference between V and $f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_\rho) \cup \{0\}$ (cf. [10, Lemma 8.1]).

Corollary 4.11. *Let $d \in \mathbb{N}_{\geq 2} \cup \{\infty\}$, and let $(\mathbf{b} = (b_j)_{j=1}^d, \mathbf{n} = (n_j)_{j=1}^d)$ be a pair of weight and frequency data such that for all $i = 2, \dots, d$, the set $\{\ln n_1, \ln n_i\}$ is linearly dependent over $\mathbb{Q}$. Let V denote the multiplier variety associated with $K_{\mathbf{b},\mathbf{n}}$. Suppose that $\sigma = \sigma_{\mathbf{a}}(\sum_{j=1}^d b_j n_j^{-s})$.*

- (i) *If $\sum_{j=1}^d b_j n_j^{-\sigma} \geq 1$, then $V = f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_{\frac{\rho}{2}}) \cup \{0\}$, where ρ is the unique real number satisfying $\sum_{j=1}^d b_j n_j^{-\rho} = 1$.*
- (ii) *If instead, $\sum_{j=1}^d b_j n_j^{-\sigma} < 1$, then $V = f_{\mathbf{b},\mathbf{n}}(\overline{\mathbb{H}_{\frac{\rho}{2}}}) \cup \{0\}$.*

Proof. (i) Suppose that $\sum_{j=1}^d b_j n_j^{-\sigma} \geq 1$. Then, $K_{\mathbf{b},\mathbf{n}}$ is well defined on $\mathbb{H}_{\frac{\rho}{2}}$. Hence, by the definition of multiplier variety, $f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_{\frac{\rho}{2}}) \cup \{0\} \subseteq V$. Thus, it remains to show that $V \subseteq f_{\mathbf{b},\mathbf{n}}(\mathbb{H}_{\frac{\rho}{2}}) \cup \{0\}$. To this end, let $0 \neq \mathbf{w} = (w_j)_{j=1}^d \in V$. Then, by Theorem 4.6, we have

$$\mathbf{w} \in \bigcap_{J \in \mathcal{J}(\mathbf{n})} Z(q_J) \cap \mathbb{B}_d. \quad (4.31)$$

By the assumption, for every integer $i = 2, \dots, d$, the set $J_i := \{1, i\} \in \mathcal{J}(\mathbf{n})$. Hence, by Proposition 4.5, there exist integers $x_i, y_i \in \mathbb{N}^2$ with $\gcd(x_i, y_i) = 1$ such that

$$n_i^{x_i} = n_1^{y_i}, \quad i = 2, \dots, d. \quad (4.32)$$

If q_{J_i} is the polynomial associated to J_i , then by (4.31) we have

$$q_{J_i}(\mathbf{w}) = b_i^{\frac{x_i}{2}} w_1^{y_i} - b_1^{\frac{y_i}{2}} w_i^{x_i} = 0, \quad i = 2, \dots, d.$$

Since $\mathbf{w} \neq 0$, it follows that $w_1 \neq 0$. Solving for w_i , we obtain

$$w_i = \sqrt{\frac{b_i}{b_1^{r_i}}} w_1^{r_i}, \quad i = 2, \dots, d, \quad (4.33)$$

Immediately after the proof, on source PDF page 24, the authors write:

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We can prove the converse of the above corollary in many particular cases, and we believe that the converse is true in general. However, a general proof remains unknown.

5. ISOMETRIC ISOMORPHISM OF MULTIPLIER ALGEBRAS

Given a subset $S \subseteq \mathbb{C}^d$, the *affine hull* of S , denoted by $\text{aff}(S)$, is defined as

$$\text{aff}(S) := \lambda + \text{span}(S - \lambda)$$

Thus the open problem is to prove the converse: does either exact curve description force (1)? We prove that it does, simultaneously for both alternatives and for finite or countably infinite data.

2 Definitions and a closedness observation

Write $\mathbb{H}_\gamma = \{s \in \mathbb{C} : \text{Re } s > \gamma\}$ and let $\mathbb{B}_d$ be the open unit ball of $\mathbb{C}^d$ when $d < \infty$, or of ℓ^2 when $d = \infty$. If $f = f_{\mathbf{b}, \mathbf{n}}$ maps $\mathbb{H}_\gamma$ into $\mathbb{B}_d$, the generated multiplier variety is

$$V = \{z \in \mathbb{B}_d : g(z) = 0 \text{ for every } g \in \mathcal{M}_d \text{ with } g|_{f(\mathbb{H}_\gamma)} = 0\}.$$

Every multiplier is holomorphic, hence norm-continuous, on $\mathbb{B}_d$. Consequently, V is relatively norm closed in $\mathbb{B}_d$ and $f(\mathbb{H}_\gamma) \subseteq V$.

3 Proof intuition

If $\log n_i / \log n_1$ is irrational, move vertically along the curve in steps of $2\pi / \log n_1$. Coordinate 1 then returns exactly to its starting phase, whereas coordinate i follows a dense irrational rotation. Choose a limit phase for coordinate i that is not among the countably many phases compatible with the fixed first coordinate. Compactness of finite phase tori—or a diagonal argument in countably many coordinates—makes all coordinates converge. In the infinite case the fixed amplitudes are square summable, so small tails turn coordinatewise convergence into norm convergence. Closedness places the limit in V , but the incompatible pair of phases proves that it is not on the Dirichlet curve. Hence exact curve equality is possible only in rational rank one.

4 The converse theorem

Theorem 1 (Norm-closure obstruction). *Let $d \in \mathbb{N}_{\geq 2} \cup \{\infty\}$, let $b_j > 0$ and $n_j \geq 2$, and suppose*

$$f(s) = (\sqrt{b_j} n_j^{-s})_{j=1}^d \quad (s \in \mathbb{H}_\gamma)$$

takes values in $\mathbb{B}_d$. Let $W \subseteq \mathbb{B}_d$ be relatively norm closed and contain $f(\mathbb{H}_\gamma)$. If

$$\alpha_i := \frac{\log n_i}{\log n_1} \notin \mathbb{Q}$$

for some $i \geq 2$, then W contains a nonzero point which is not $f(s)$ for any complex s for which $f(s)$ is defined. In particular,

$$W \neq f(\mathbb{H}_\gamma) \cup \{0\}, \quad W \neq f(\overline{\mathbb{H}_\gamma}) \cup \{0\}$$

whenever the second displayed image is defined in $\mathbb{B}_d$.

Proof. Fix a real $x > \gamma$ and set

$$a_j = \sqrt{b_j} n_j^{-x} > 0.$$

Then $a = (a_j) \in \mathbb{B}_d$. The subgroup

$$G_i = \{e^{-2\pi i m \alpha_i} : m \in \mathbb{Z}\}$$

is countable and dense in $\mathbb{T}$: density is the elementary irrational-rotation theorem (equivalently, the closure of an infinite cyclic subgroup of $\mathbb{T}$ is all of $\mathbb{T}$). Choose $\zeta \in \mathbb{T} \setminus G_i$, and choose distinct integers k_r such that

$$e^{-2\pi i k_r \alpha_i} \longrightarrow \zeta. \quad (2)$$

For

$$s_r = x + \frac{2\pi i k_r}{\log n_1}$$

we have

$$f(s_r)_j = a_j e^{-2\pi i k_r \alpha_j}, \quad \alpha_j = \frac{\log n_j}{\log n_1}. \quad (3)$$

In particular, the first phase is always 1, while the i -th phase tends to ζ .

If $d < \infty$, compactness of $\mathbb{T}^d$ gives a subsequence on which every phase in (3) converges. If $d = \infty$, successive extraction followed by the diagonal subsequence gives the same conclusion coordinate by coordinate. Write the limiting phases as $\eta_j \in \mathbb{T}$. We retain

$$\eta_1 = 1, \quad \eta_i = \zeta, \quad (4)$$

and define $w = (a_j \eta_j)_{j=1}^d$.

For completeness, the convergence is in norm even when $d = \infty$. Since $f(x) \in \mathbb{B}_\infty$,

$$\sum_{j=1}^{\infty} a_j^2 = \sum_{j=1}^{\infty} b_j n_j^{-2x} < 1.$$

Given $\varepsilon > 0$, first choose J so that $4 \sum_{j>J} a_j^2 < \varepsilon^2/2$. Coordinatewise convergence makes

$$\sum_{j \leq J} a_j^2 \left| e^{-2\pi i k_r \alpha_j} - \eta_j \right|^2 < \varepsilon^2/2$$

for all sufficiently large r , while the remaining sum is at most $4 \sum_{j>J} a_j^2$. Hence $f(s_r) \rightarrow w$ in ℓ^2 . The finite-dimensional case is immediate. Moreover,

$$\|w\|^2 = \sum_j a_j^2 = \|f(x)\|^2 < 1,$$

so $w \in \mathbb{B}_d$. Since every $f(s_r)$ belongs to W and W is relatively norm closed, $w \in W$. Also $w \neq 0$ because $w_1 = a_1 > 0$.

It remains to exclude w from the curve. Suppose $w = f(s)$, and write $s = u + iv$. Comparing the modulus of coordinate 1 gives

$$\sqrt{b_1} n_1^{-u} = |w_1| = a_1 = \sqrt{b_1} n_1^{-x},$$

so $u = x$. Comparing its phase and using $w_1 = a_1 > 0$ gives $e^{-iv \log n_1} = 1$, hence

$$v = \frac{2\pi m}{\log n_1} \quad \text{for some } m \in \mathbb{Z}.$$

The phase of coordinate i would therefore be $e^{-iv \log n_i} = e^{-2\pi i m \alpha_i} \in G_i$. But (4) says that this phase is $\zeta \notin G_i$, a contradiction. Thus w has all the asserted properties. $\square$

Corollary 2 (Full converse to Corollary 4.11). *Under the hypotheses and notation of Corollary 4.11 of [1], each equality in that corollary is an if-and-only-if statement. Explicitly, in either of its alternatives the displayed formula for V holds exactly when*

$$\frac{\log n_i}{\log n_1} \in \mathbb{Q} \quad (i = 2, \dots, d).$$

Equivalently, the $\mathbb{Q}$ -linear span of $\{\log n_j : 1 \leq j \leq d\}$ has dimension one.

Proof. The forward direction from rational rank one to the appropriate equality is Corollary 4.11 of [1]. Conversely, apply Theorem 1 with $W = V$. The multiplier variety is relatively norm closed and contains the relevant open-half-plane image. If any displayed ratio were irrational, the theorem would produce a nonzero point of V outside even the entire Dirichlet curve, contradicting either the open-image equality in alternative (i) or the closed-image equality in alternative (ii). $\square$

5 Verification, scope, and novelty

The proof uses no polynomial presentation of V and no compactness claim for the infinite-dimensional ball. The only compactness is coordinatewise phase compactness, followed by the displayed square-summable-tail estimate. The constructed limit stays strictly inside the ball, which is essential because V is only relatively closed there.

The result answers precisely the conjecture after Corollary 4.11; it does not address the paper's broader classification questions. A bounded literature check through 12 August 2026 searched the current arXiv record and v3 full text, the exact title, arXiv id, exact conjecture sentence, author publication pages, citation listings, and close multiplier-variety/Dirichlet-curve keywords. No later solution or claim of this converse was found. This is a bounded novelty screen, not an exhaustive bibliographic proof of novelty.

Human-review recommendation. High priority. Check the relative norm-closedness observation, the $d = \infty$ tail estimate, and the final two-coordinate phase contradiction.

References

- [1] H. Ahmed, B. K. Das, and C. K. Sahu, *Multiplier varieties and multiplier algebras of CNP Dirichlet series kernels*, arXiv:2507.21537v3 [math.FA], 14 May 2026.